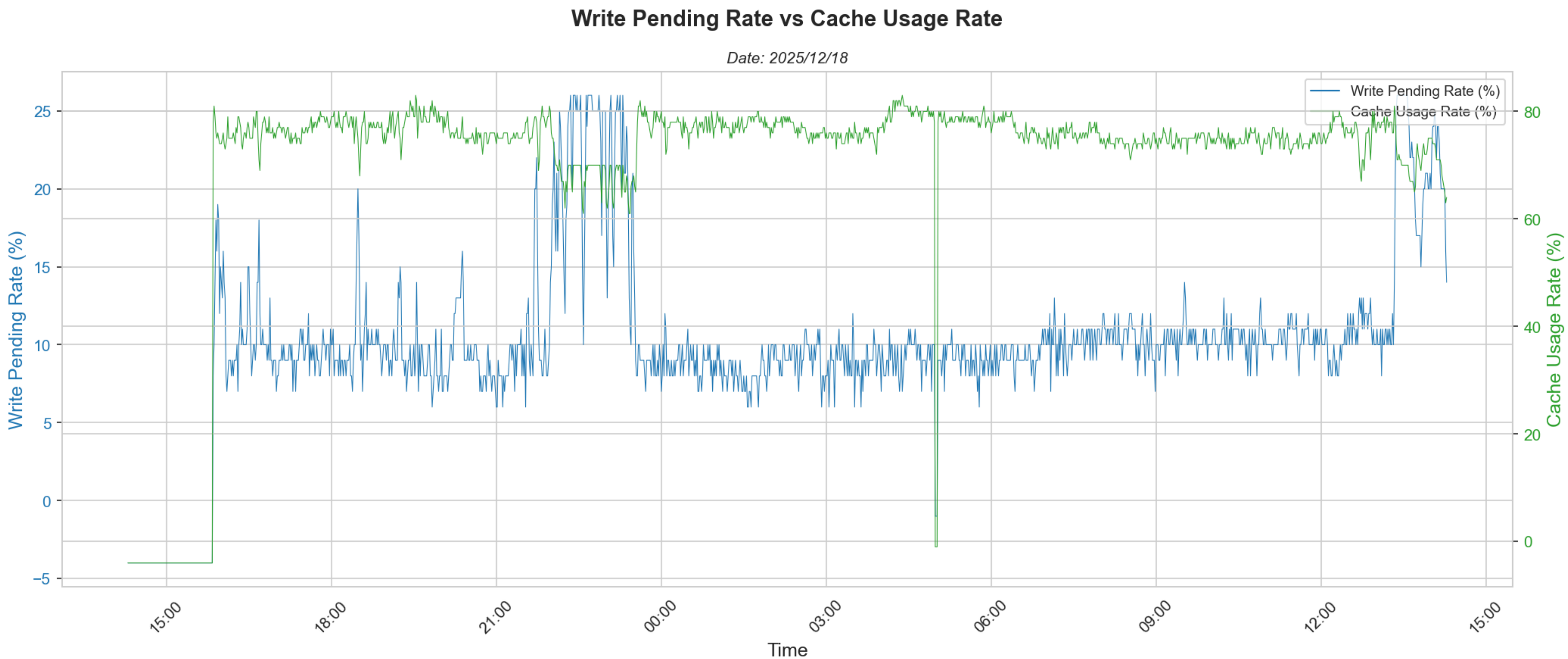
VSP5600 Performance Analysis Report

Storage Performance & Capacity Analysis

Report Generated: 20260108_085004
Data Extraction Date: 20260108_085004

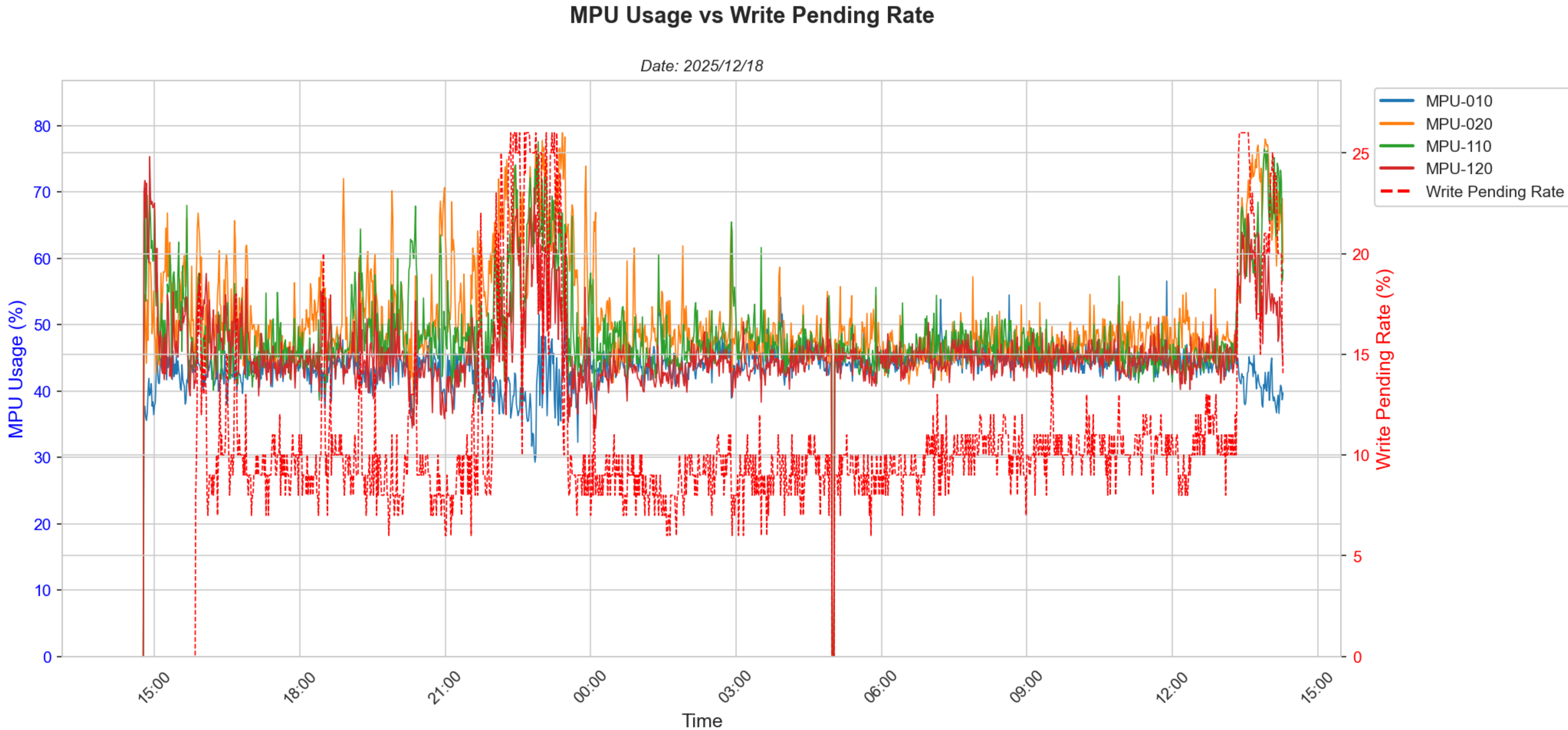
This report provides a comprehensive analysis of storage performance metrics collected from the Hitachi VSP 5600 system. It includes detailed charts and analysis of cache performance, processor utilization, port response times, and LDEV (Logical Device) metrics.

6.1 Cache Metrics: Write Pending Rate vs Cache Usage Rate



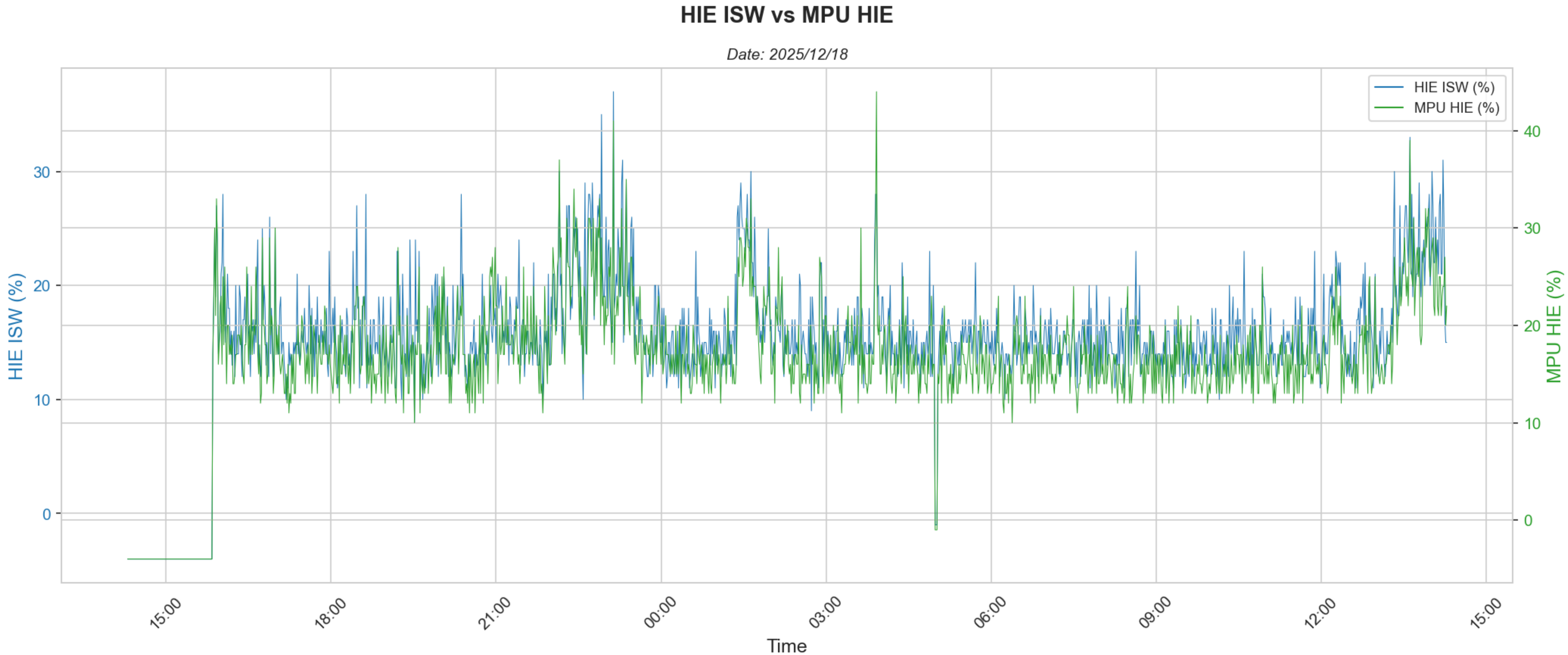
Analysis:
Cache write pending is associated with high transfer rates on LDEVs. It also causes high MPU usage as the MPs will be busy trying to destage the write data that is coming in. In addition cache usage for read data cache goes down to accomodate for the write data coming in. This may cause higher read response times.

6.2 MPU Usage Analysis



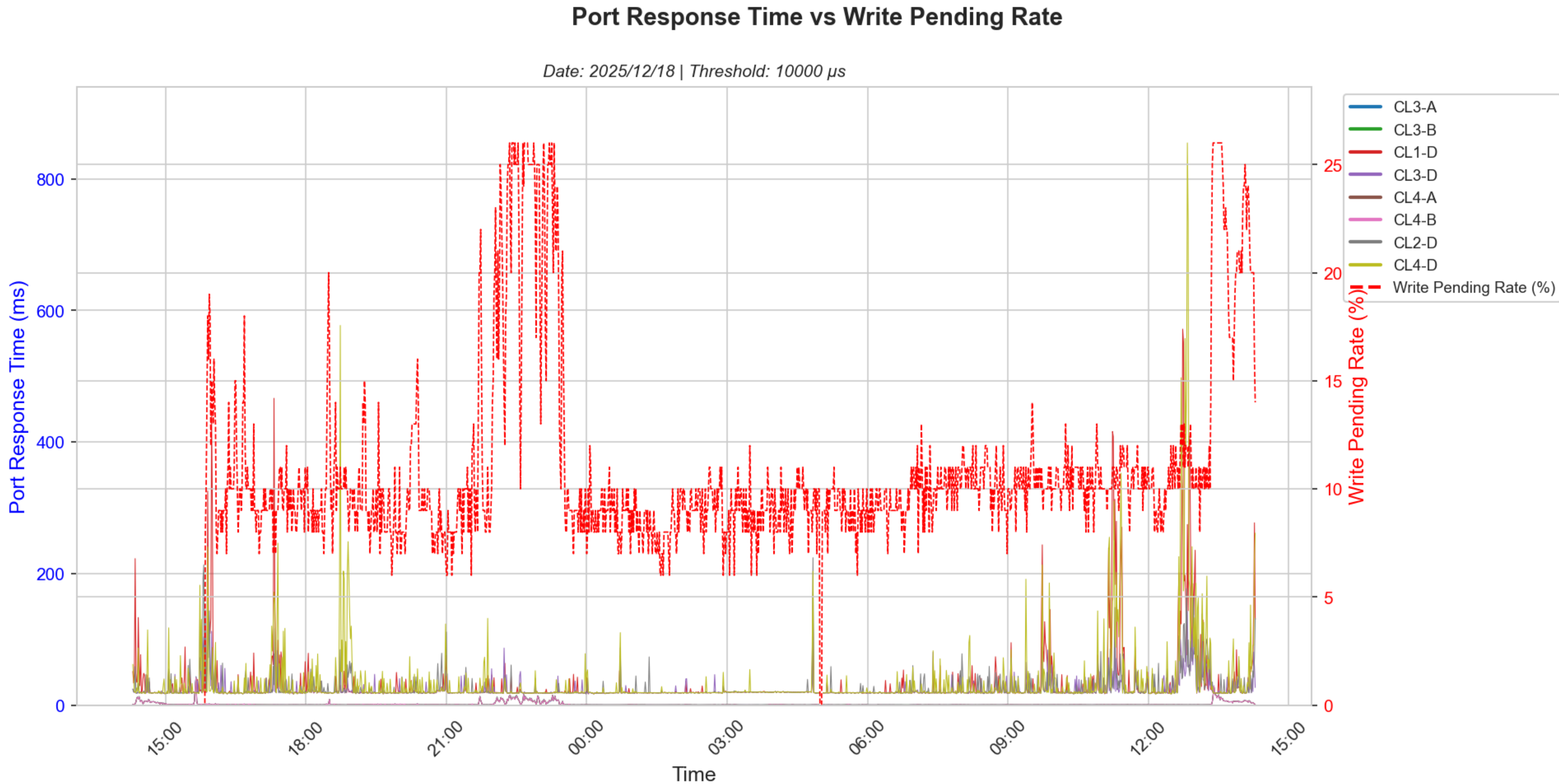
Analysis:
MPU usage can be high owing to high write transfer, high IOPS or background tasks such as UR Journal processing, ShadowImage processing or Thin Image operations.

6.3 HIE Metrics: Internal Path Utilization



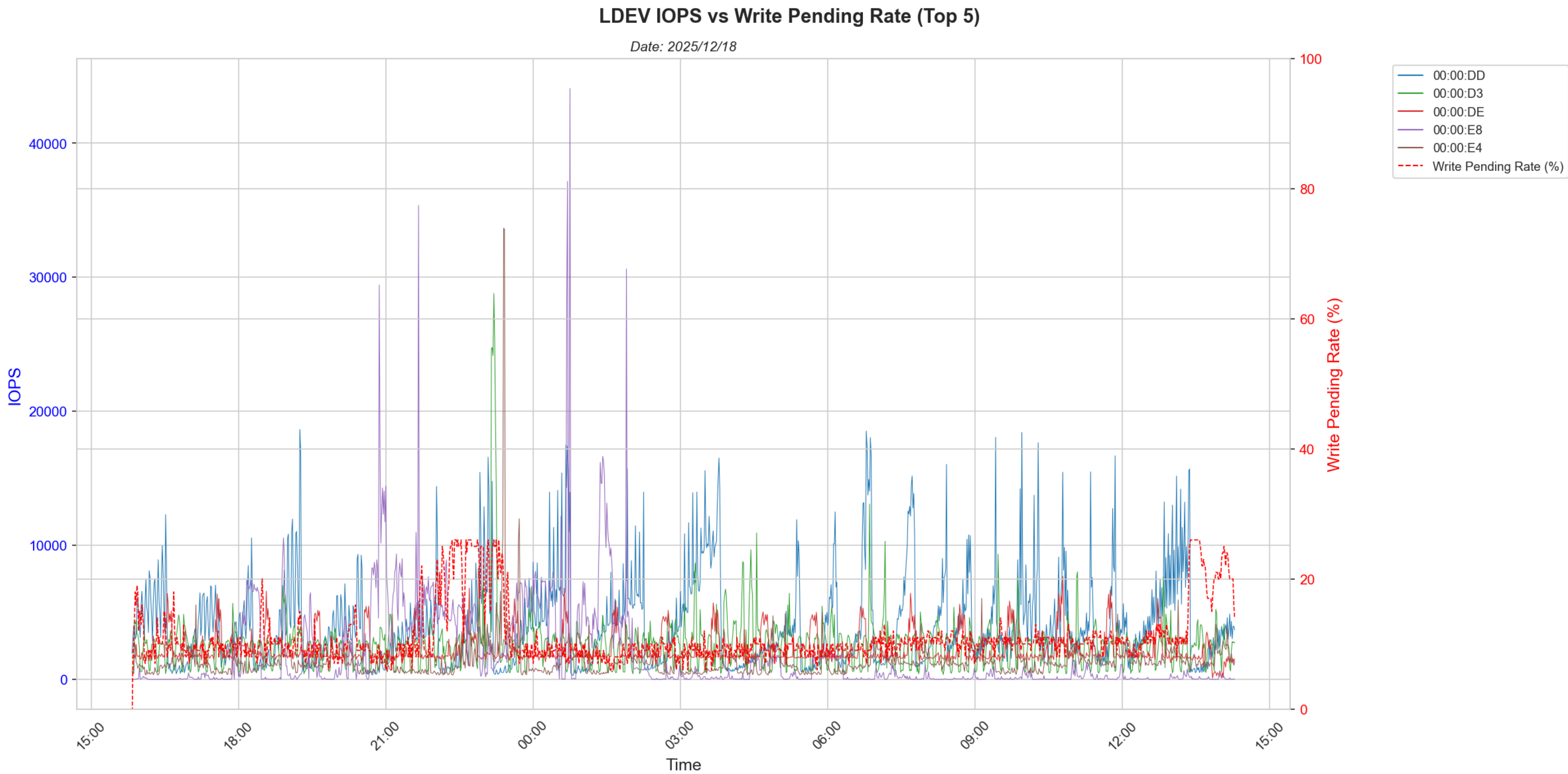
Analysis:
HIE - ISW and MPU - HIE internal path traffic should not be more than 40% used. At around 40% or more LDEV response times may get affected.

6.4 Port Response Times



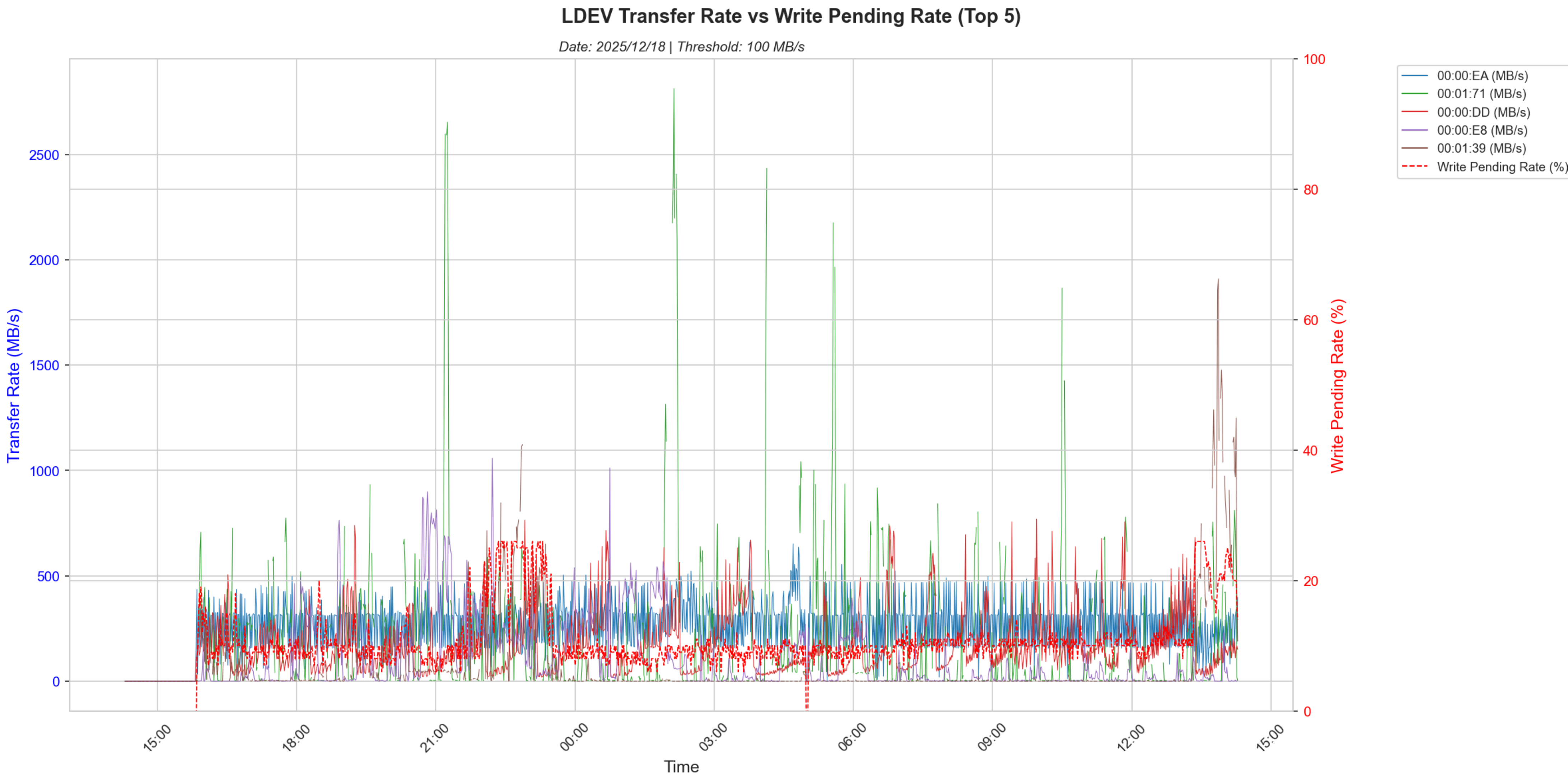
Analysis:
Port response times are normally high owing to port saturation i.e. high transfer rates or high IOPS. They are also correlated with write pending as high transfer rates may cause high write pending.

6.5 LDEV IOPS Analysis



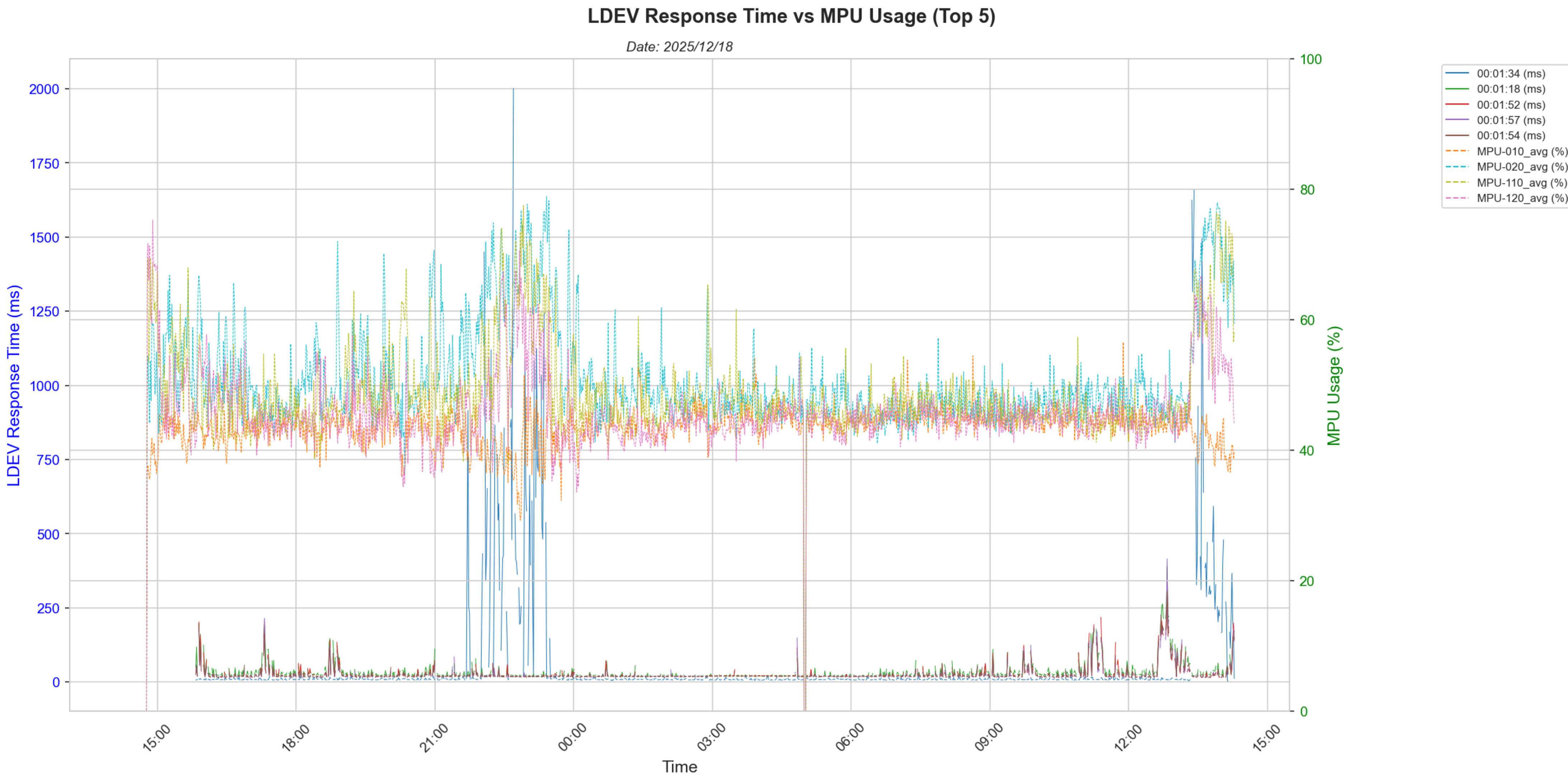
Analysis:
LDEVs with high IO activity are identified via a adaptive threshold filter and the top five LDEVs are identified in terms of IOPS. These LDEVs could be causing noisy neighbour issues i.e. hogging all the DP Pool bandwidth, thus starving the other LDEVs of IO bandwidth.

6.6 LDEV Transfer Rate Analysis



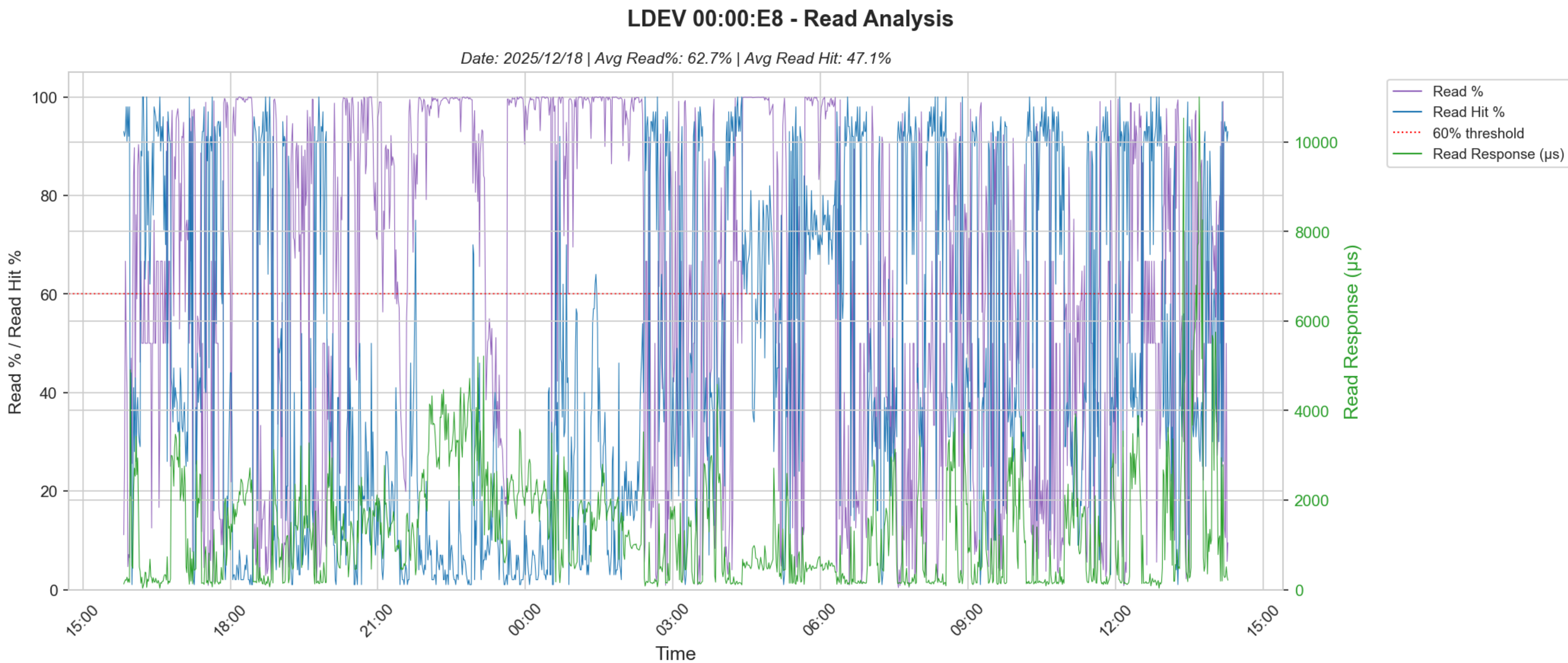
Analysis:
LDEVs with high transfer rates are typically sequential. Note that the LDEVs identified here may not be the same as the LDEVs identified in Section 6.5. These LDEVs are also identified using a adaptive threshold filter.

6.7 LDEV Response Time vs MPU Usage



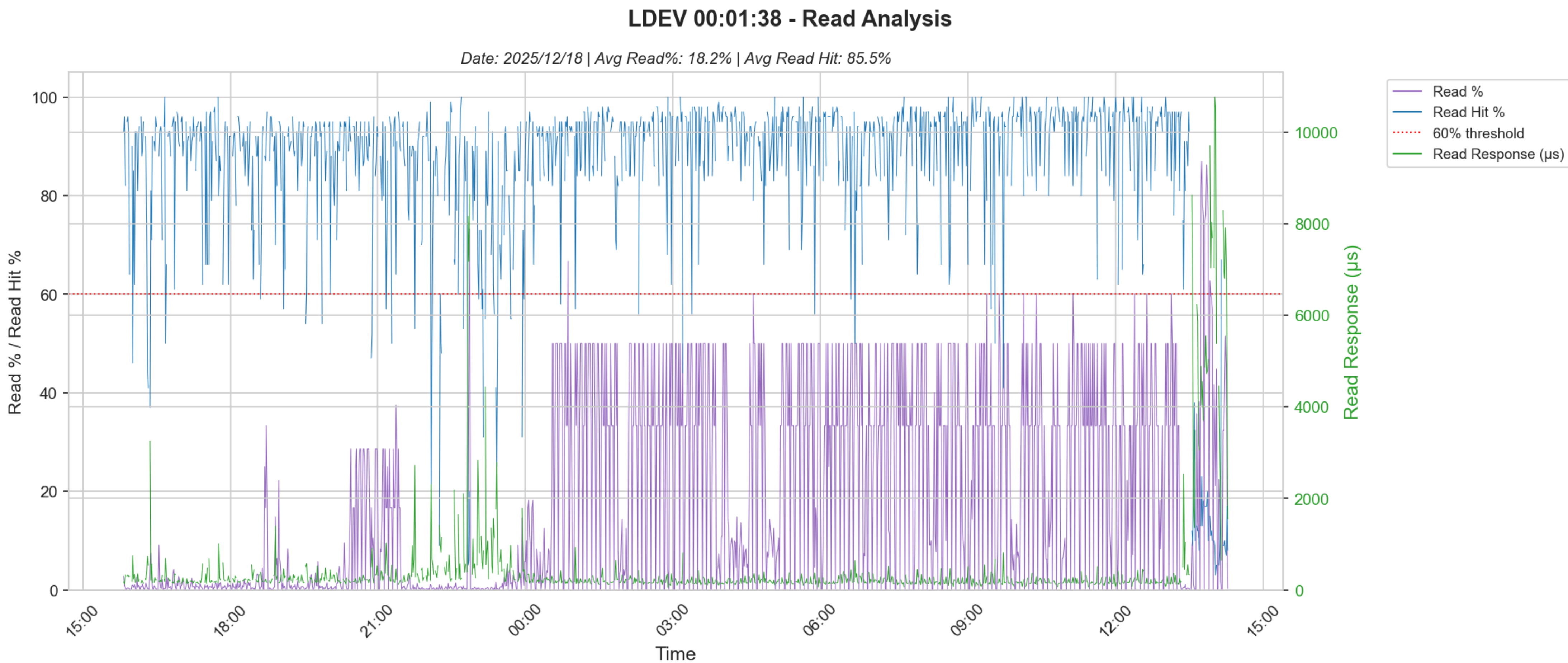
Analysis:
Here the LDEVs that are affected by high write pending and by high MPU usage are identified. These may not be the same the ones identified earlier. They could be LDEVs that are IO starved.

6.8 LDEV Read Hit Analysis - LDEV 01:00:00:E8X



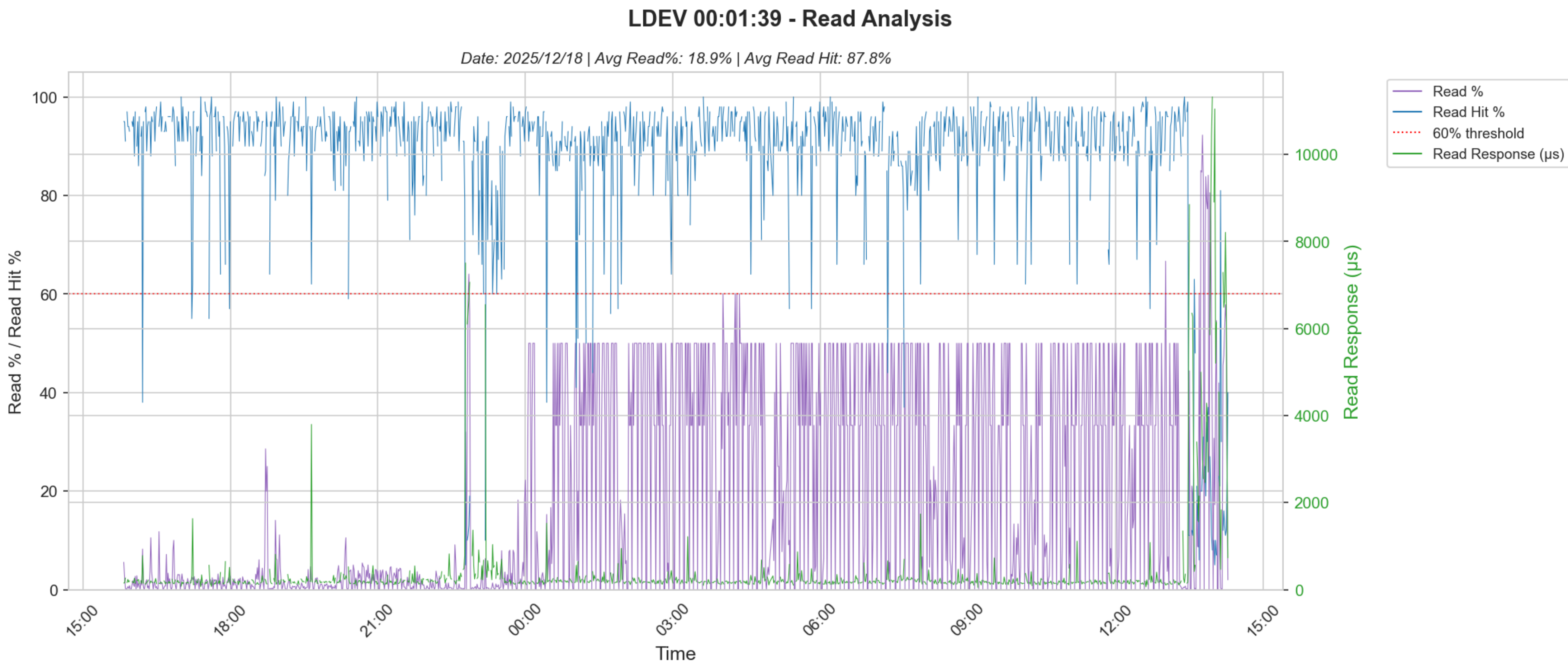
Analysis:
LDEVs with low read hits, high read percentage and high read response times are identified here. Higher read response times and low read hits could be because of a noisy neighbour.

6.8 LDEV Read Hit Analysis - LDEV 02:00:01:38X



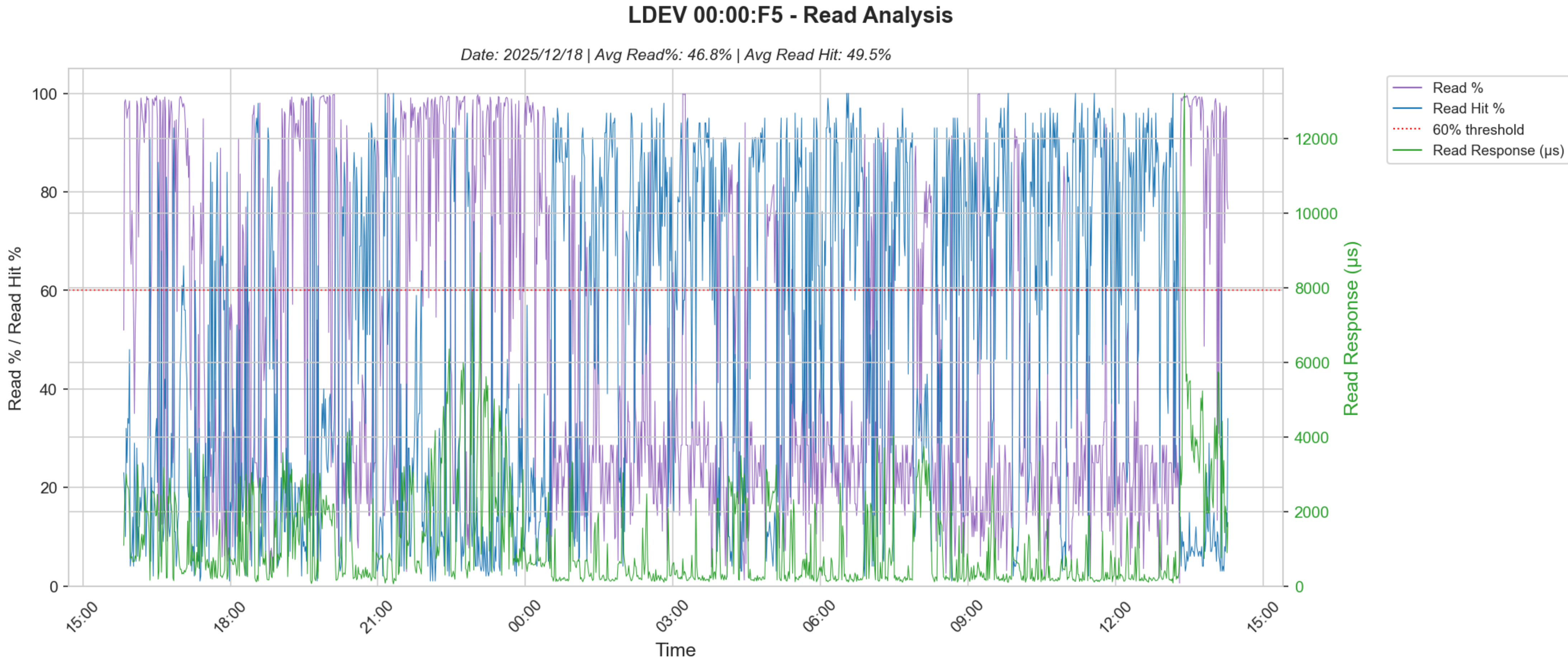
Analysis:
LDEVs with low read hits, high read percentage and high read response times are identified here. Higher read response times and low read hits could be because of a noisy neighbour.

6.8 LDEV Read Hit Analysis - LDEV 03:00:01:39X



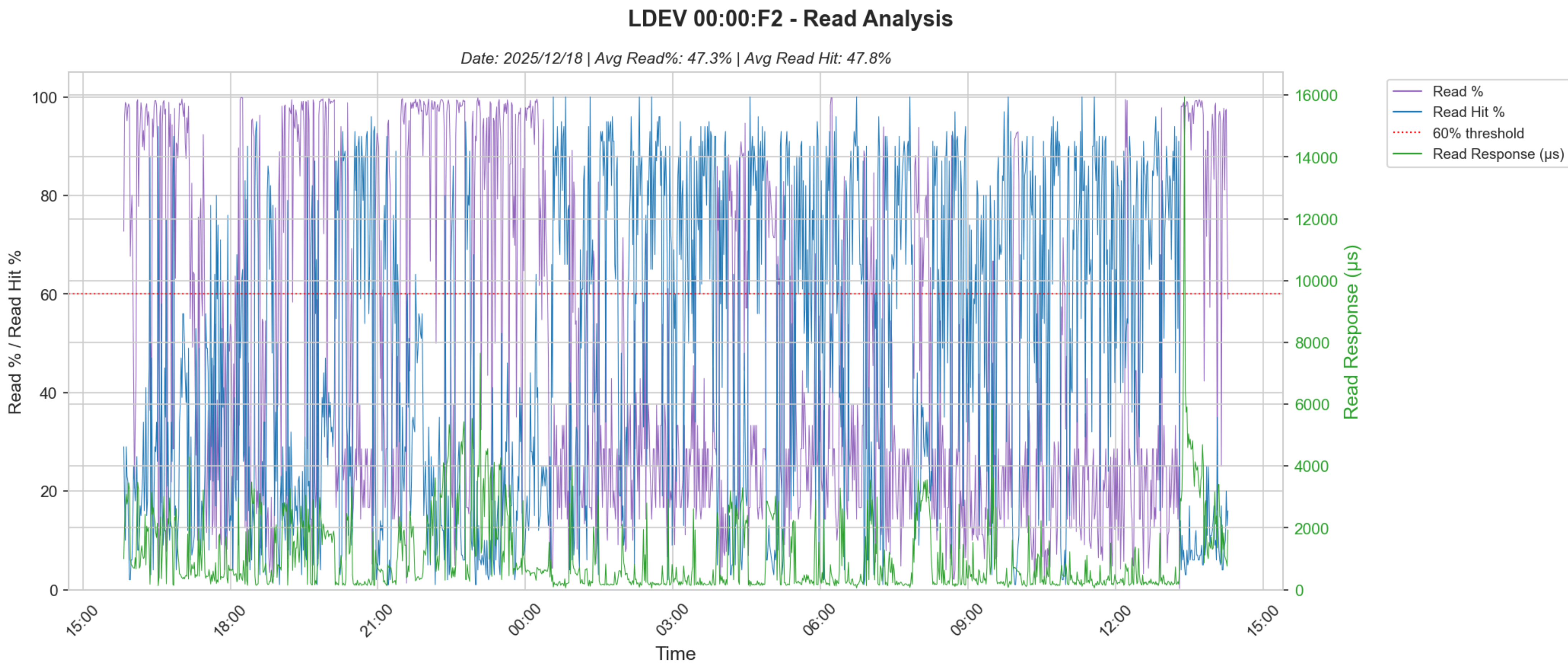
Analysis:
LDEVs with low read hits, high read percentage and high read response times are identified here. Higher read response times and low read hits could be because of a noisy neighbour.

6.8 LDEV Read Hit Analysis - LDEV 04:00:00:F5X



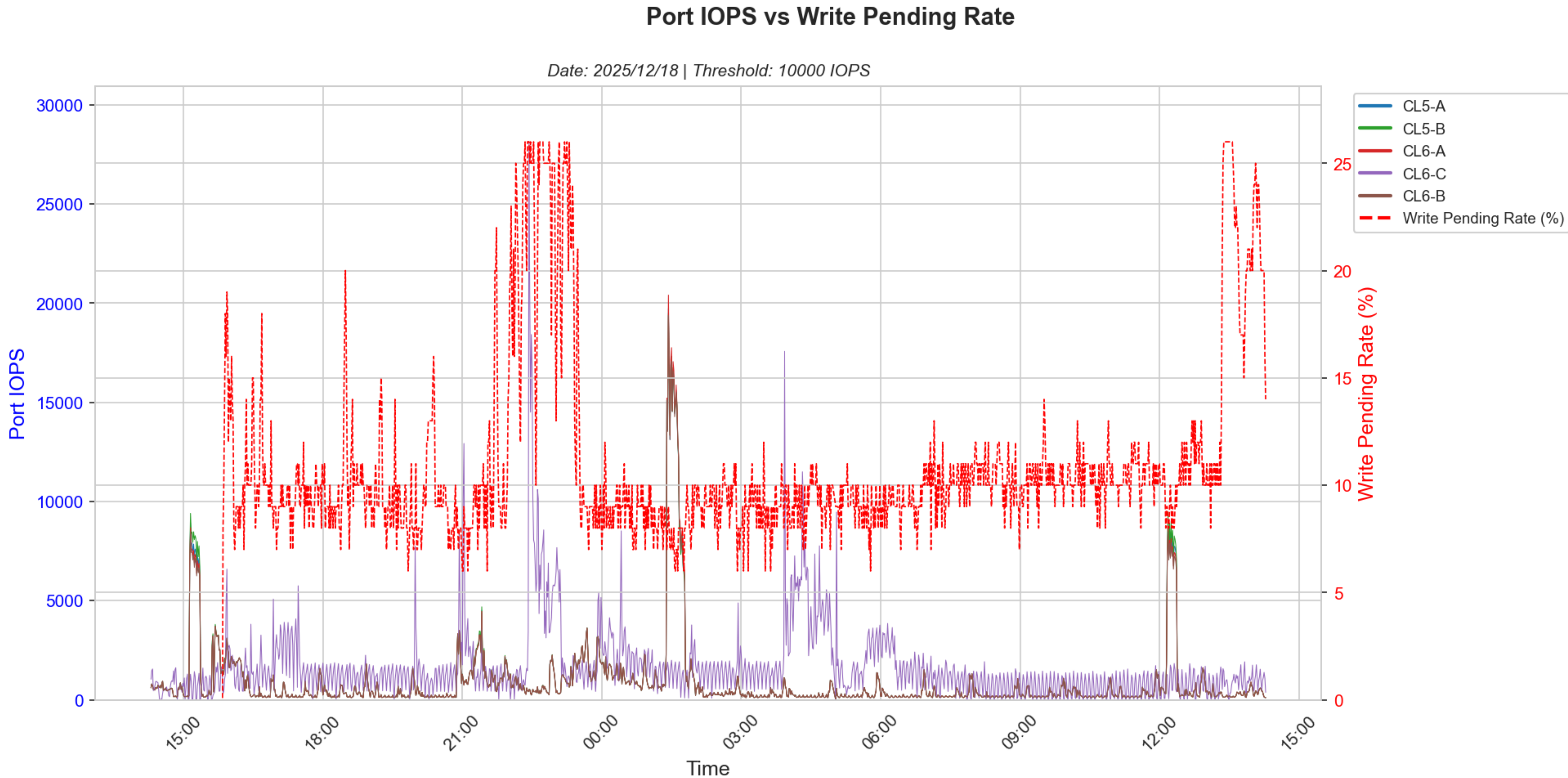
Analysis:
LDEVs with low read hits, high read percentage and high read response times are identified here. Higher read response times and low read hits could be because of a noisy neighbour.

6.8 LDEV Read Hit Analysis - LDEV 05:00:00:F2X



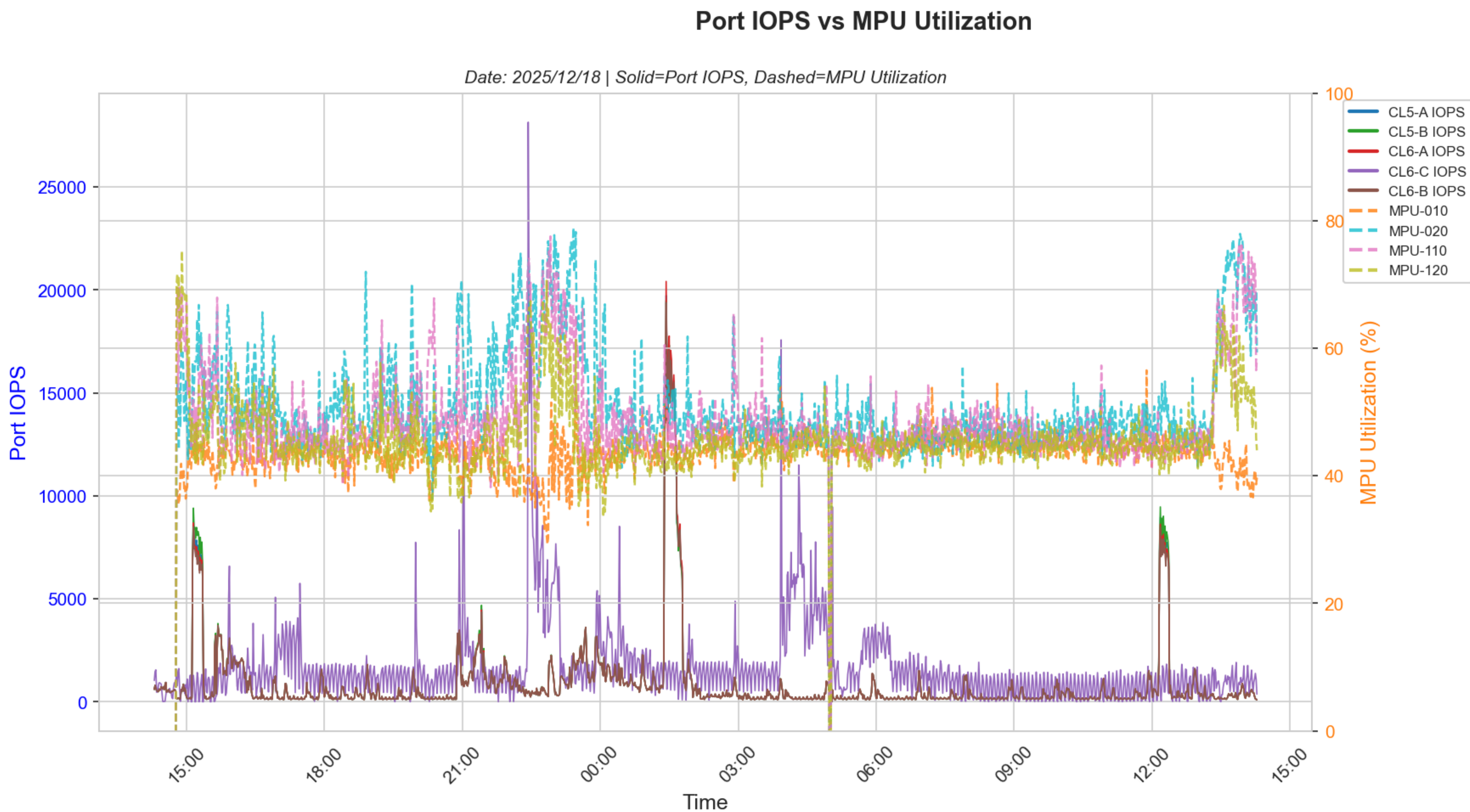
Analysis:
LDEVs with low read hits, high read percentage and high read response times are identified here. Higher read response times and low read hits could be because of a noisy neighbour.

6.9 Port IOPS Analysis



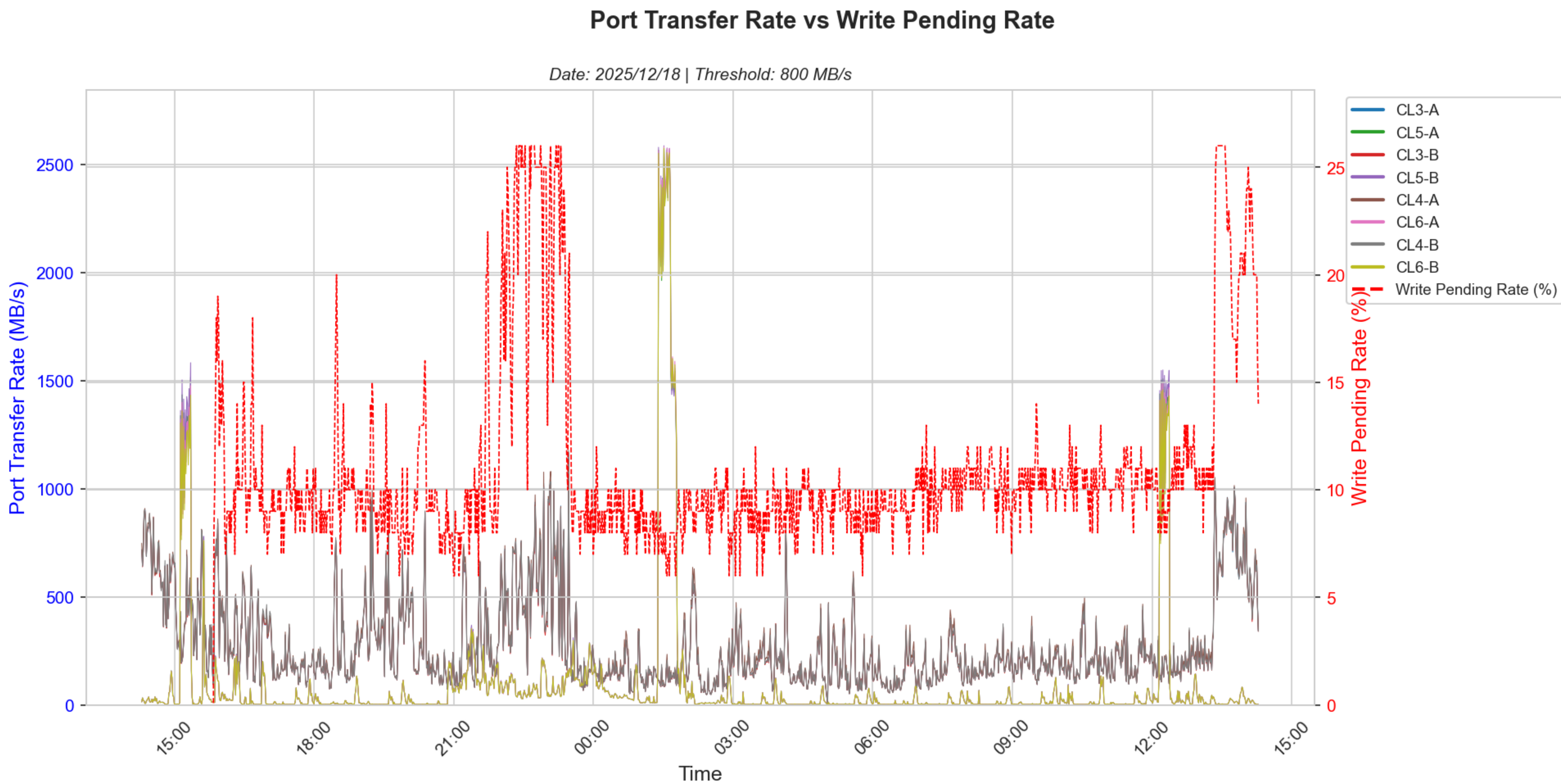
Analysis:
High port IOPS are typically small block random IO. This tends to push MP busy rates or cause write pending issues if mostly write IOPS.

6.9 Port IOPS vs MPU Utilization



Analysis:
This chart correlates Port IOPS with MPU Utilization. High port IOPS combined with elevated MPU utilization may indicate processor bottlenecks. If MPU utilization remains low despite high IOPS, the system has adequate processing headroom.

6.10 Port Transfer Rate Analysis



Analysis:
High transfer rate ports will cause LDEV response time issues if the ports are saturated. They will also cause high write pending if the transfer is mostly writes.